

NOT-INVENTED-HERE SYNDROME AND INNOVATION PERFORMANCE: THE CONFOUNDING EFFECT OF INNOVATION CAPABILITIES AS ORGANISATIONAL ROUTINES IN SERVICE FIRMS

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Currently, it is commonly accepted that the not-invented-here syndrome (NIHS) refers to internal resistance against external knowledge, and that it has devastating consequences on innovation performance (IP). Nevertheless, studies on organisational practices that neutralise NIHS are incipient. Moreover, it is accepted that innovation capabilities (ICs) are perhaps the organisational routines with the highest positive influence on IP. In this study, based on the organisational routine framework and agency theory, we propose a model in which ICs (i.e., client-focused, marketing-focused, and technology-focused ICs) have a confounding effect on an NIHS–IP relation. Structural equation modelling (SEM) was used to test the proposed model with survey data from a sample of 161 service firms. We find that (1) NIHS influences IP negatively, and (2) ICs weaken NIHS–IP relation, practically turning it null. Finally, this paper provides insights for researchers and managers regarding a possible overestimation of the negative effects of NIHS on the literature.

Keywords: Not-invented-here syndrome; innovation performance; innovation capabilities; confounding effect; service innovation; emerging economies.

Introduction

In the last 30 years, the literature has presented the not-invented-here syndrome (NIHS) as a phenomenon of apocalyptic proportions which has devastating consequences on an innovation process and mainly on its results. Hence, for many years, there has been a wave of studies specifically focused on establishing all its characteristics and consequences (Lichtenthaler and Ernst, 2006; Antons and Piller, 2015). Just recently, empirical studies have been undertaken which have opened the possibility of neutralising this attitude to reject knowledge coming from external sources. This has contributed to demystifying the destructive action of NIHS on an innovation process. For instance, it has been possible to show that professional training reduces the effect of NIHS on certain aspects of innovation performance (IP) (De Araújo *et al.*, 2014); nevertheless, in this case, it has to do with a very specific organisational practice that has restrictions as time, mode and place.

On the other hand, innovation capabilities (ICs) are probably the organisational routines that have the most positive influence on IP (Yam *et al.*, 2011; Martinez-Roman *et al.*, 2011; Sok and O'Cass, 2011). Therefore, IP is found simultaneously under the negative effects of NIHS and the positive effects of ICs. However, there is a notable lack of studies that simultaneously analyse these antagonist relations, despite the fact that there are two study perspectives in the literature. The first one explores the incidence of certain attitudes in the formation of ICs (Rothaermel and Hess, 2007; Lin, 2007). By contrast, the second one states that organisational capacities can mold attitudes (Zbaracki and Bergen, 2010). This is precisely the approach that has been recently adopted and which has opened the possibility of neutralising NIHS in organisations (De Araújo *et al.*, 2014). However, there are no studies which specifically explore the relationship between an organisational capacity, such as ICs, and a concrete attitude, such as NIHS, related to the influence of both constructs on IP.

Therefore, in the understanding that NIHS can be modified, we propose in this article that ICs, understood as organisational routines, can neutralise an attitude such as NIHS (Feldman and Pentland, 2003). Likewise, based on the Agency Theory (Eisenhardt, 1989; Wright *et al.*, 2001), we also consider that the positive effect of ICs on IP can reduce the individuals' opportunistic behaviours underlying NIHS. Thus, the consequences of NIHS on IP will be reduced drastically or definitively.

To test the proposed model, we conducted a confounding effect analysis using structural equation modelling (SEM) and taking a partial least squares (PLS) approach. This type of analysis has been developed primarily in the context of health sciences and epidemiological research, and more recently in psychological studies. A confounder is defined as a "variable that changes the relation of an

independent variable because it is related to both the independent and the dependent variable” (Meinert, 1986, p. 285). The confounding hypothesis suggests that a third variable explains and particularly obscures the relationship between an independent variable and a dependent variable (MacKinnon *et al.*, 2000; Mehio-Sibai *et al.*, 2005; Li *et al.*, 2007; MacKinnon, 2008; Coffman and Zhong, 2012). Unfortunately, the word confounding has been used synonymously with several other terms, and has also been used to refer to at least four distinct concepts (Greenland and Morgenstern, 2001). In this study, ICs are the confounding factor that obscures an NIHS–IP relation. We used a recently developed scale for measuring this construct in service firms. (Hogan *et al.*, 2011).

We provide three main contributions to the literature: (1) we offer empirical evidence regarding the neutralisation of the effects of NIHS on IP by means of the action of an organisational routine such as ICs in service companies in an emerging country which is a technology follower. Service companies are either knowledge-intensive services (KIS) or less knowledge-intensive services (LKIS) (2) we address NIHS based on Agency Theory and an organisational routine framework. Hence, we expand analysis possibilities and we offer greater understanding of this phenomenon. (3) we use a confounding-effect model which is widely used in other disciplines to propose a new analysis alternative which could be of great use in future studies in the field of business research. In this field, there is saturation and an excessive use of mediation and moderating effects models. The latter has insuperable multicollinearity problems (Little *et al.*, 2006; Echambadi and Hess, 2007; Dalal and Zickar, 2012; Wurm *et al.*, 2014).

The rest of this paper is organized in sections. Section 2 presents the theoretical background and hypotheses development of this study. Section 3 describes the research methodology employed. Section 4 presents data analysis and results, and Sec. 5 shows the discussion and implications of results.

Theoretical Background and Hypotheses

‘NIHS is conceived as the tendency of a project group of stable composition to believe it possesses the monopoly of knowledge of its field, which leads it to reject new ideas from outsiders to the detriment of its performance’ (Katz and Allen, 1982, p. 7). Its main symptom is a strong emphasis on intra-team innovation relations and communications (De Araújo *et al.*, 2014). This attitude of irrational devaluation and rejection of knowledge inputs which may be of great value to a company has various explanations related to NIHS functions in organisational members. The main functions are ego defensive, value-expressive, social adjustive, knowledge and utilitarian (Antons and Piller, 2015).

The first three functions are closely related to each other, and refer to employees' needs to define, express and defend their self-identity, self-concepts and values, and maintain their social relations (Mehrwald, 1999; Gupta and Govindarajan, 2000; Hussinger and Wastyn, 2015; Rubera *et al.*, 2016). By contrast, a knowledge function is a confirmation of a bias; it gives more importance to capturing information from a setting which does not deviate from each individual's experience and opinions (Knudsen and von Zedtwitz, 2003). Meanwhile, a utilitarian function is related to individuals' opportunism particularly regarding their interests in obtaining benefits and avoiding situations that jeopardise their personal interest (Bohner and Dickel, 2011).

The above brings to light the complexity and extent of the irrationality of NIHS which leads employees to a conscious or unconscious rejection of external knowledge: This rejection is bigger when this knowledge comes from competitors (Hussinger and Wastyn, 2015). Consequentially, this attitude has multiple effects on various organisational variables and mainly on IP (Tranekjer and Knudsen, 2012). IP is understood as tangible outputs in terms of generation of new upgraded products and processes (Sok and O'Cass, 2011; Alegre *et al.*, 2006; Hagedoorn and Cloodt, 2003). Oddly, an NIHS seminal study tried to explain the decline of R&D team results which over time have lost effectiveness in attaining their innovation results (Katz and Allen, 1982).

Specifically, NIHS has a negative effect on IP, for it favours a mistaken evaluation of external technologies and their suboptimal use (Lichtenthaler and Ernst, 2006). This leads companies to not acquire these technologies timely, even though they can represent a source of competitive advantage and the generation of radical innovations, or to implement them slowly or even implement them with flaws in business processes and in product development. This situation also causes reprocesses and delays in the innovation process (Katz and Allen, 1982) and innovation team demotivation (Menon and Pfeffer, 2003), which in the long run lead to the decline of innovation results.

Additionally, rejection of external knowledge deteriorates companies' IP, since it prevents them from identifying new business ideas which arise from the combination of internal and external knowledge (Lichtenthaler and Ernst, 2006). Furthermore, NIHS constitutes a barrier to exchanging knowledge with the environment (Tranekjer and Knudsen, 2012), especially when deploying open innovation practices (De Araújo *et al.*, 2014), which allow reaching technological complementarities and economies of scale in developing new products with external partners (Dahlander and Gann, 2010; Huizingh, 2011). Therefore, this study hypothesizes:

H1: *NIHS influences IP negatively.*

It has been amply demonstrated that IP is influenced positively by ICs (Yam *et al.*, 2011; Martinez-Roman *et al.*, 2011; Sok and O'Cass, 2011). ICs are a type of organisational ability or routine related to the generation of new ideas and knowledge in order to take advantage of market opportunities (Elmqvist and Le Masson, 2009; Camisón and Villar-López, 2014; Yam *et al.*, 2004). Nevertheless, studies which have been conducted to analyse ICs in service companies are recent (Den Hertog *et al.*, 2010; Kindström *et al.*, 2013; Plattfaute *et al.*, 2015; Chen *et al.*, 2016). An outstanding one is Hogan *et al.*'s work (2011), in which the authors identify the main ICs for these types of companies and develop a measure scale with good reliability and validity indicators.

According to Hogan *et al.* (2011), service ICs are of three types: (1) client-focused IC (CFIC), a company's ability to provide clients with services and products that offer unique benefits superior to those of competitors, and to solve clients' problems in innovative ways; (2) marketing-focused IC (MFIC), the ability to develop and implement novel promotional approaches, and innovative marketing programs; and (3) technology-focused IC (TFIC), an ability to adopt new software, integrated systems and technology, and to innovate with new software and technology in order to keep ahead of the market.

At the same time, ICs are closely related to employees' behaviours and attitudes. In the literature, there are various studies that have been conducted seeking to describe how behaviours and attitudes affect the formation of ICs (Subramaniam and Youndt, 2005; Assink, 2006; Rothaermel and Hess, 2007; Lin, 2007). In this perspective, the effects of attitudes represented in NIHS regarding innovation processes have been partially addressed (Tranekjer and Knudsen, 2012; De Araújo *et al.*, 2014).

In this paper, we propose, though, that there is an opposite relation, which is that ICs affect NIHS. This is stated based on the theory of dynamic capabilities (Teece *et al.*, 1997) and Agency theory (Eisenhardt, 1989). To develop this statement, it is first necessary to point out that ICs are a type of dynamic capability (Wang and Ahmed, 2007), which is understood as a learned and stable pattern of collective activity (Zollo and Winter, 2002). In other words, dynamic capabilities are organisational routines by which firms achieve new resources configurations as markets emerge, collide, split, evolve and die (Eisenhardt and Martin, 2000; Wollersheim *et al.*, 2013).

More specifically, organisational routines are composed of ostensive and performative aspects (Feldman, 2000). The former is the ideal or schematic form of a routine. The latter consists of specific actions, by specific people, in specific places and times that bring routine to life (Feldman and Pentland, 2003). These two aspects turn a routine not only into a source of stability, but also into a source of

change because people react in many ways towards the results of the previous iterations of a routine (Pentland and Feldman, 2008).

In this regard, Feldman and Pentland (2003) propose that particularly the ostensive aspect of a routine affects employees significantly because it shows which types of attitudes or behaviours are deemed legitimate or are part of a recognisable routine. Therefore, these authors consider that the creation and enforcement of organisational routines constitute a domination or coordination mechanism (Becker, 2004) because it allows certain control of the decisions employees make while at work (Braverman, 1974; Miller *et al.*, 2012). This situation can form the locus for on-going dynamics of domination and resistance (Pentland and Feldman, 2005). For this, ICs can totally weaken or annul a negative NIHS–IP relation. This happens because ICs exert certain domination of employees, leading them to adopt innovating behaviours related to a greater degree of opening to external knowledge.

Nevertheless, simultaneously, routines also favour the rise of truces which constitute a “stable accommodation” (Nelson and Winter, 1982, p. 108) between employees’ motivations and organisational needs; routines are able to mold the way employees perceive and describe their own interests (Zbaracki and Bergen, 2010). These truces imply that an employee’s cognitive conflicts cease, as well as their emotional conflicts (Pentland and Feldman, 2005; Becker *et al.*, 2005). Now then, truces are temporary because at certain times they collapse, rise, and are established again, and this situation occurs repetitively (Zbaracki and Bergen, 2010; Hodgkinson and Healey, 2014).

Based on the above, ICs as routines can encourage the rise of truces or “stable accommodations” between rejection and the use of external knowledge. A truce rises between NIHS and the ostensive aspect of innovation routines (Chesbrough, 2006; Laursen and Salter, 2006). This means that ICs exert pressure and mold employee perception of outside knowledge up to the point of sometimes aligning it with the organisational aims, demanding a greater use of this resource. That is to say, ICs manage to reduce this irrational rejection that NIHS supposes. Therefore, there is neutralisation of NIHS and its effects on IP.

These employees’ attitude and behavioural changes derived from organisational routines can also be analysed based on the postulates of Agency Theory (Eisenhardt, 1989). The theory mainly proposes that employees permanently act with opportunism, which is perceived as self-interest seeking with guile (Williamson, 1975), as is the case for NIHS. Nevertheless, the studies that have tried to take Agency Theory beyond its traditional limits have found that employees can set aside opportunism to act cooperatively and generate optimal outcomes. This is because of a need for personal achievement at work, because of social norms of obligation and reciprocity and because employees’ self-interests are cooperatively interrelated (Wright *et al.*, 2001; Fontrodona and Sison, 2006;

Cuevas-Rodríguez *et al.*, 2012). Therefore, when IP is improved by the actions of ICs, the need for achievement in employees' innovation work is satisfied. As a result, their willingness to act without bias towards outside knowledge is increased, because they consider that the use of that knowledge does not jeopardise their personal interest. On the contrary, it becomes a key resource to help maintain motivation at work. Besides, improvements on IP indicators result in multiple stimuli and benefits — economic or otherwise — for employees who satisfy their own interests. This allows a reduction of their opportunistic behaviours, such as NIHS, and favours their willingness to use outside knowledge because this activity generates personal benefits.

Based on the above, this article proposes that ICs can totally weaken or annul a negative NIHS–IP relation. ICs invigorate innovation results; that is, they affect IP positively, and thus, counter the negative effect of NIHS. At the same time, ICs as organisational routines mold employees' attitudes and reduce their resistance to the use of knowledge from external sources. Therefore this study hypothesises:

H2: *ICs have a confounding effect on an NIHS–IP relation.*

Methodology

Sample and data collection

Figure 1 illustrates the research model for this study. This was tested with data from a sample of service companies from Colombia, an emerging country

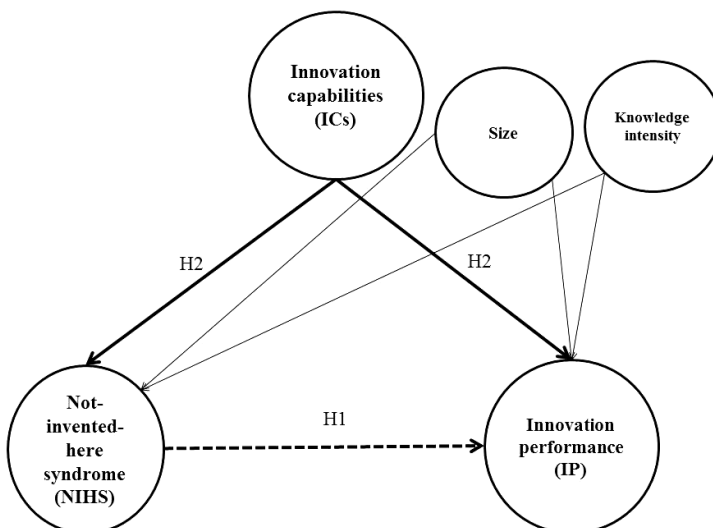


Fig. 1. Research model.

and technological follower (Hoskisson *et al.*, 2000; Castellacci, 2011). This sector represents 60% of the Colombian GDP, generates about 47% of the country's jobs and participates with 1% of the nation's overall exports. Another particularity of the companies in this sector is that most of the innovation they generate is not new to the world; it is just new to the company and the local market. Moreover, investment in R&D, which represents only 0.2% of the GDP, is mainly conducted by the Government, and company participation in it does not surpass 40%.

In Colombia, it can be very difficult for scholars to collect survey data directly from companies. To facilitate our data collection, we hired the services of a specialised mailing company which sent the forms to executives in service companies located all throughout the country. We conducted fieldwork from June to July 2014, and we received a total of 161 valid questionnaires. If we assume a

Table 1. Sample characteristics and key informants' job posts.

	Frequency	Percentage
Characteristic		
Professional, scientific and technical activities	48	30
Wholesale and retail trade	36	22
Information and communication	24	15
Human health and social work activities	16	10
Education	11	7
Financial and insurance activities	4	2
Electricity and gas supply	4	2
Administrative and support service activities	6	4
Accommodation and food service activities	3	2
Transportation and storage	2	1
Real estate activities	2	1
Arts, entertainment and recreation	1	1
Other service activities	4	2
Company size (number of employees)		
Small and medium	107	66
Large	54	34
Key informant position		
Chief executive officer/General manager	101	63
Marketing manager	16	10
I+D+i manager	9	6
IT manager	8	5
Financial managers	6	4
Human resource managers	3	2
Other executives	18	11

medium effect size, the proposed model would require a minimum sample size of 91 to achieve a power of 0.8 and an alpha level of 0.05 (Cohen, 1988). In our study our sample size is 161, which suggests a satisfactory effect size. Table 1 shows detailed information on sample characteristics and informants' jobs.

Measures

ICs were measured using a scale developed by Hogan *et al.* (2011) for service companies, which includes 13 items (see Appendix A). These items were applied using the stem, "Compared to competitors, our company's ability to... is" on a Likert scale that ranged from "much worse than competitors", to "much better than competitors". As for measuring NIHS, we employed three items from the scale developed by Mehrwald (1999) which are commonly adapted in several studies (Tranekjer and Knudsen, 2012; De Araujo *et al.*, 2015), and have been measured on a Likert scale from "totally disagree" to "totally agree" (see Appendix A). To measure IP, we used Sok and O'Cass (2011). As with the ICs items, respondents evaluated their company's IP relative to their main competitors on a Likert scale that ranged from "much worse than competitors", to "much better than competitors" (see Appendix A).

Data Analysis and Results

Common method bias

To avoid concerns of common method bias (CMB), we adopted three strategies. First, we assured study respondent anonymity and confidentiality; second, we decided to mix the order of the questions using different scale types; and finally, we performed a Harman's one-factor test as a post hoc statistical procedure to check CMB (Podsakoff *et al.*, 2003; Chang *et al.*, 2010). The variance explained by the first primary component was only 47.6%, indicating that CMB is not a likely contaminant of the results.

Reliability and validity

The reflective measurement model was analysed using SmartPLS 2.0 to examine its reliability and validity (Henseler *et al.*, 2009). Individual item reliability is adequate when an item has a factor loading that is greater than 0.7 for its construct or dimension (Carmines and Zeller, 1979). In this study, reflective indicators satisfy this requirement, except in the case of indicator CFIC1 (see Table 2). This item remains in the study on account of its contribution to content validity (Chin,

1998; Roldán and Sánchez-Franco, 2012). In addition, we evaluate the construct reliability by checking whether the construct composite reliability (CR) and indicator loadings are above the threshold of 0.7 (Chin, 2010). Moreover, we test convergent validity by examining whether the construct average variance extracted (AVE) is above the recommended value of 0.5 (Fornell and Larcker, 1981; Hair *et al.*, 2014). According to the values in Table 2, the above condition is met in all cases.

As for discriminant validity, this is evaluated using one of the most commonly used criteria: the square root value of the AVE for each construct should be greater than the inter-correlations between constructs in the model (Chin, 1998). According to the values in Table 3, the above condition is met in all cases.

Table 2. Reliability and convergent validity.

Constructs	SL	SE	<i>t</i> -value	CR	AVE
Innovation capabilities				0.94	0.56
CFIC1	0.61*	0.08	7.92		
CFIC2	0.79*	0.04	18.61		
CFIC3	0.80*	0.04	20.66		
CFIC4	0.80*	0.04	20.27		
CFIC5	0.71*	0.05	13.43		
MKIC1	0.70*	0.06	11.02		
MKIC2	0.68*	0.07	9.88		
MKIC3	0.73*	0.06	12.96		
MKIC4	0.78*	0.05	14.72		
TFIC1	0.74*	0.06	13.38		
TFIC2	0.78*	0.05	14.86		
TFIC3	0.82*	0.05	17.90		
TFIC4	0.75*	0.06	12.38		
Not-invented-here syndrome				0.81	0.59
NIHS1	0.70*	0.14	5.14		
NIHS2	0.81*	0.12	6.77		
NIHS3	0.79*	0.13	6.17		
Innovation performance				0.93	0.81
IP1	0.93*	0.02	53.58		
IP2	0.93*	0.02	57.61		
IP3	0.84*	0.05	15.26		
Knowledge intensity	1.00	0.00	0.00	1.00	1.00
Size	1.00	0.00	0.00	1.00	1.00

Note: SL, standardised loading; SE, standard error; CR, composite reliability; AVE, average variance extracted.

* $p \leq 0.001$.

Table 3. Discriminant validity analysis.

Factors	1	2	3
1. IP	0.90		
2. ICs	0.54	0.75	
3. NIHS	-0.26	-0.32	0.77

Note: The diagonal elements are the square root of the AVE; off-diagonal elements are the correlations among constructs.

Control variables

The study includes controls for contextual variables that might potentially confound the results. Previous studies show that organisation size and knowledge intensity influence ICs and IP in firms (Camisón-Zornoza *et al.*, 2004; Paladino, 2008; Wu, 2012; Denicolai *et al.*, 2014). We add the following control variables:

- Size dummy: Small and medium firms as 0 and large companies as 1.
- Knowledge intensity dummy: LKIS as 0 and KIS as 1.

Confounding effect testing

Following the approach by Mehio-Sibai *et al.* (2005), Li *et al.* (2007) and Coffman and Zhong (2012), we conducted a test on the confounding effect of ICs by assessing a confounded model and a direct model using PLS and using bootstrapping (5000 resamples) to generate standard errors and *t*-statistics, in order to evaluate the statistical significance of the path coefficients (Henseler *et al.*, 2009). Table 4 accounts for the results of this test for both the direct and the confounded model, comparing the fit between them. The results provide evidence that NIHS has a negative effect on IP ($\beta = -0.28$, *t*-value = 2.86). Thus, we can accept H1. In the confounded model, the path between ICs and IP is significant with a positive sign ($\beta = 0.51$, *t*-value = 6.27), and a path between ICs and NIHS is also significant but with a negative sign ($\beta = -0.30$, *t*-value = 2.70). However, the path between NIHS and IP in the confounded model is negative but not significant ($\beta = -0.11$, *t*-value = 1.23). Thus, the direct effect in the confounded model lost strength becoming even non-significant. Therefore, we can accept H2. With regard to the control variables, size and knowledge intensity are not significant either.

Predictability and predictive relevance of both models

The model predictability is evaluated by R^2 values for the dependent latent variables. Table 4 shows that all the endogenous constructs have a R^2 value that

Table 4. Results of the structural models.

Models	Structural path	St. coefficients	t-value	Conclusion
Direct	<i>Direct effects</i>			
	NIHS→ IP ($R^2 = 0,08$; $Q^2 = 0,01$)	-0.28*	2.86	H1. supported
	<i>Control variables</i>			
	Size→ IP	0.06	0.60	
	Knowledge intensity→ IP	-0.02	0.20	
Confounded	<i>Direct effects</i>			
	ICs→ NIHS ($R^2 = 0,12$; $Q^2 = 0,04$)	-0.30*	2.70	H2. supported
	ICs→ IP	0.51**	6.27	
	NIHS→ IP ($R^2 = 0,30$; $Q^2 = 0,19$)	-0.11	1.23	
	<i>Control variables</i>			
	Size→ IP	0.02	0.27	
	Size→ NIHS	-0.01	0.10	
	Knowledge intensity→ IP	-0.07	0.76	
	Knowledge intensity→ NIHS	-0.12	1.05	

* $p \leq 0.05$; ** $p \leq 0.001$.

exceeds the minimum value of 0.1 (Falk and Miller, 1992), except for the IP in the direct model which is very close to this value. Moreover, the R^2 value for IP in the confounded model indicates that the theoretical model proposed explains 30% of the variance of the construct, which is a very satisfactory level of predictability. In the case of the model's predictive relevance, this is evaluated by Q^2 values for the dependent latent variables. Table 4 shows that all the constructs have a Q^2 value greater than zero (Hair et al., 2014). Therefore, both models have satisfactory predictive relevance.

Discussion and Implications

Discussions of the results

Once again this study proves what has been demonstrated in the literature, regarding the positive influence of ICs on IP and the negative influence of NIHS. Nevertheless, the most important finding is related to the role ICs play in softening employees' attitudes; particularly, those related to the rejection of external knowledge, and in the long run, in neutralising NIHS. Based on the above, it is logical to claim that in the literature they could be overestimating the negative effects of NIHS on innovation results. This occurs because the syndrome is not analysed in connection with organisational routines, in this case innovation, when

in reality employees' attitudes are conditioned by the molding effect of the ostensive aspect of routines. This is clearly evidenced in the results, given that once ICs are incorporated in the analysis, the NIHS-IP relation which at first was negative and significant will become practically null. Moreover, the confounded model has more predictability and predictive relevance.

Hence, the results demonstrate that ICs weaken an NIHS-IP relation because they have a positive effect on innovation results. This softens the opportunistic behaviours of employees who can obtain personal benefits by satisfying the need for accomplishment and personal interest. ICs also neutralise NIHS since the ostensive aspect of these types of routines acts as a mechanism of coordination, and even of domination, which also generates truces between employees' motivations and the organisational objectives regarding innovation.

Theoretical and methodological implications

This paper contributes to the field of study of innovation management in several directions. It contributes to the development of an emerging line of study which aims to describe how NIHS can be modified based on organisational practices and routines (De Araújo and Fosfuri, 2015; De Araújo *et al.*, 2014). This is of great importance if we take into account that the literature has been mainly interested in identifying the background and the consequences of NIHS (Lichtenthaler and Ernst, 2006; Antons and Piller, 2015), and in explaining the formation of ICs based on employees' attitudes and behaviors (Rothaermel and Hess, 2007; Lin, 2007; Hodgkinson and Healey, 2011).

Additionally, the horizons of the analysis of NIHS have been significantly expanded since we began to address in depth an approach based on different theoretical perspectives, such as the theory of social identity, of attitude and of ego-defense mechanisms, among others (Antons and Piller, 2015). In this regard, our work contributes to the field of study in that it incorporates Agency Theory and organisational routines in the study of NIHS. Both have great explanatory power which in a complementary way contribute to understanding the effects of NIHS, and the circumstances in which they can be modified.

Also, the study contributes empirical evidence related to the consequences of NIHS and the impact of ICs on IP in service companies, including KIS and LKIS. This is of great importance for the discipline since traditionally studies which have addressed these two issues have been mainly conducted in the manufacturing sector. Furthermore, the analysis of the data allowed us to evidence that the scale recently developed to measure ICs in service companies (Hogan *et al.*, 2011) presents good reliability and validity indicators. Therefore, its use is recommendable in studies carried out in this context which incorporate this construct.

Another contribution of the study is that it provides evidence in service companies of an emerging country which is also a technological follower.

The most important contribution is perhaps a methodological approach related to the use of the confounding-effect model to explain the relation between constructs, which is habitual enough in the context of health sciences and epidemiological research. Hence, the field of study can be boosted methodologically because a different way of conceiving relations between variables is incorporated. This goes beyond mediation and moderation effects models, which have been practically become the only ways to consider third-party influence variables in the field of technology and innovation management.

Practical implications

The presence of NIHS, which usually generates alarm and concern among innovation managers, can be countered in the short-term with an IC development strategy. In this way firms can obtain early innovation results. These results can be quickly translated into incentives that satisfy employees' own interests and enable an increase of the acceptance of social behaviours aimed at the capture and use of external information on market and technology. Thus, there can be a neutralisation of the opportunistic behaviour that NIHS supposes. New values and beliefs among people who associate the search for personal interest with the use of information from external sources can also be introduced.

Limitations and directions for future research

The main limitation of the study is the scale to measure NIHS, which is used reiteratively in the literature that partially reflects all the theoretical development the construct has had recently. For instance, it could be quite enriching to count on items that ask more specifically for NIHS manifestations related to its five functions: ego defensive, value-expressive, social adjustive, knowledge and utilitarian. Therefore, future researchers should concentrate on developing a new scale to measure NIHS with greater content validity.

This study has only explored the effect of ICs as innovation routines on an attitude such as NIHS. Nevertheless, from a theoretical point of view, there is a reciprocal influence between the ostensive and performative aspects of routines (Pentland and Feldman, 2008). In other words, routines mold employees, and thus, they constitute a source of stability; but at the same time, employees can modify routines and thus, this constitutes a source of change. Hence, future researchers should conduct empirical studies, preferably longitudinal, to establish if there is a bidirectional ICs–NIHS relation.

Appendix A. Scale Items

ICs	Innovation capabilities
CFIC	<i>Client-focused innovation capability</i>
CFIC1	Provide our clients with services/products that offer unique benefits superior to those of competitors.
CFIC2	Solve clients' problems in very innovative ways.
CFIC3	Provide clients with innovative ideas and solutions.
CFIC4	Present innovative solutions to our clients.
CFIC5	Seek out novel ways to tackle problems.
MKIC	<i>Marketing-focused innovation capability</i>
MKIC1	Develop "revolutionary industry" marketing programs for our services/products.
MKIC2	Adopt novel ways to market our company.
MKIC3	Innovate with our marketing programs to keep ahead of the market.
MKIC4	Implement innovative marketing programs.
TFIC	<i>Technology-focused innovation capability</i>
TFIC1	Innovate with new software.
TFIC2	Innovate with new technology.
TFIC3	Introduce new integrated systems and technology.
TFIC4	Adopt the latest technology in industry.
NIHS	Not-invented-here syndrome
NIHS1	Our employees regard the application of external knowledge as valuable as the application of knowledge generated internally (R).
NIHS2	Our employees have often received and used knowledge from external sources (R).
NIHS3	Our employees are positive towards developing new ideas, solutions and technologies for other companies (R).
IP	Innovation performance
IP1	Introduction of new products (completely new products and/or new product adaptation).
IP2	Our company's performance, compared to our major competitors, in terms of the number of successful new products (completely new products and/or new product adaptation).
IP3	Our company's performance, compared to our major competitors, in terms of the time to market new products (completely new products and/or new product adaptation).

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